

Simulation and Execution of Power Flows Using DIgSILENT PowerFactory and PyPower

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Abstract—This work performs power-flow studies on the IEEE 39-bus test system under different conditions and with different computational tools. Three scenarios are considered: the electric power system in its normal state, a load increase at bus 8, and a proposed solution to the effects caused by that increase. The final sections compare the results across the three scenarios and contrast the normal-state scenario with the results obtained using PyPower.

Index Terms—Power flow, IEEE 39-bus system, DIgSILENT PowerFactory, PyPower, distributed generation, static VAR compensator.

I. INTRODUCTION

Power-flow analysis is a fundamental tool for the study, planning, and operation of electric power systems (EPS), since it determines the steady-state bus voltages, voltage angles, generator dispatch, and line loadings for a given operating condition. This work performs power-flow studies on the IEEE 39-bus test system using two computational tools, DIgSILENT PowerFactory and PyPower.

Three scenarios are analyzed: the system in its normal state (Scenario 1), a load increase at bus 8 (Scenario 2), and a proposed solution to the resulting problems (Scenario 3). The results are then compared across the three scenarios, and the normal-state results are contrasted with those obtained in PyPower.

II. SCENARIO DEVELOPMENT AND SIMULATION

A. Scenario 1: 39-Bus System in Normal State and Configuration

The EPS was first assembled in PowerFactory, entering the data of its components—buses, generators, lines, transformers, and loads—from the reference document. The only elements requiring prior calculation were the transmission lines, since PowerFactory requires their parameters in nominal units (ohms and siemens); these calculations are reported in the accompanying spreadsheet (sheet “Line calculations”). With all data entered, the power flow was executed, and the results are given in the accompanying spreadsheet (sheet “PF results S”).

Fig. 1 shows the resulting bus voltages. The values remain within acceptable ranges, with bus 36 showing a slight over-voltage of 1.064 p.u., followed by bus 25 at 1.058 p.u. Fig. 2 shows the voltage angle at each bus. The remaining results are presented in the scenario-comparison section and are also available in the accompanying spreadsheet.

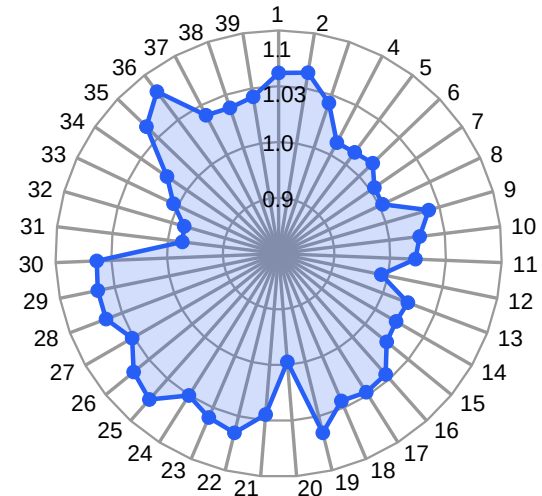


Fig. 1. Bus voltages of the EPS – Scenario 1.

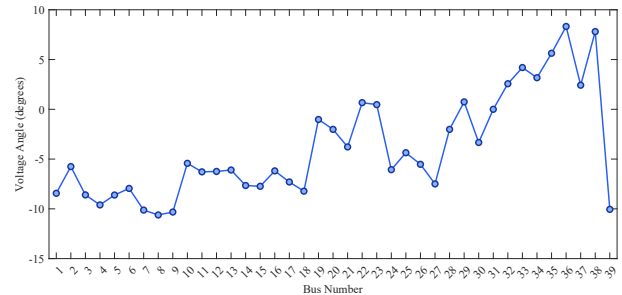


Fig. 2. Voltage angles at the EPS buses – Scenario 1.

B. Scenario 2: Load Increase at Bus 8

This scenario keeps the same line parameters, generator dispatch, and voltage data; the only change is that the load at bus 8 increases to 1305 MW and 440 MVAR. Under this new condition, the goal is to analyze the power flow and observe how the system adapts.

Connecting a large load causes a voltage drop in the surrounding area; if this drop exceeds a threshold, it is considered poor power quality, which the engineer must avoid. A margin of 6% variation from the ideal voltage (1 p.u.) is adopted for this and the next scenario. The accompanying spreadsheet

(sheet “PF results C”) reports the results with the new load and no corrective action.

Fig. 3 shows the voltage profile across the 39 buses together with the established limits. Buses 7 and 8 fall outside the limits, at 0.919 and 0.913 p.u. respectively, while buses 5 and 12 are near the limit at 0.94 and 0.943 p.u. Fig. 4 shows that the voltage angles now range between -30° and 0° , indicating that a larger angular opening was needed to meet the new conditions.

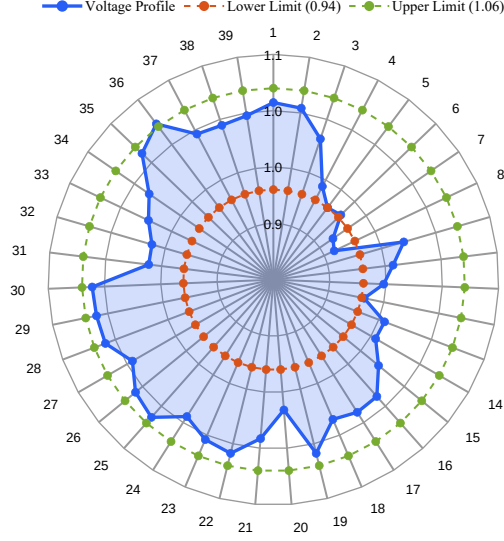


Fig. 3. Bus voltages of the EPS – Scenario 2.

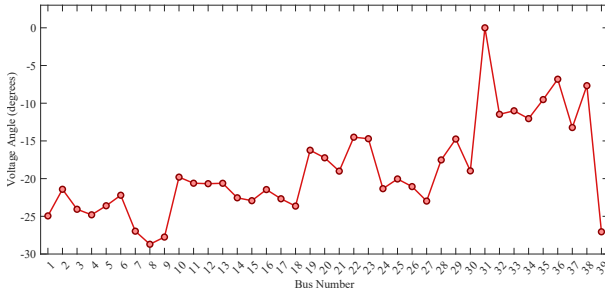


Fig. 4. Voltage angles at the EPS buses – Scenario 2.

C. Scenario 3: Improvement to Scenario 2

Large loads are unavoidable in power systems, so engineering seeks solutions such as automatic tap-changing transformers, capacitor banks, or compensation systems. With the 6% voltage threshold established, the load increase caused considerable voltage drops at nearby buses and overloaded the slack-bus generator beyond its operating limits. The proposed solution combines distributed generation (DG) and a static VAR compensator (SVS): the DG offloads the reference and other generators by helping to meet the new demand, while the SVS raises the voltage of the most-affected bus to an acceptable value.

According to [1], [2], [3], distributed generation produces energy near the consumption points, complementing centralized generation and encompassing various sources, including renewables; its benefits include reduced transmission losses, resource optimization, and improved reliability and efficiency. DG acts as a reduction in net load, easing congestion and optimizing the existing infrastructure. However, its penetration must be studied beforehand, since excessive DG can cause overloads and adverse changes in the power flows that affect voltage and frequency stability; adequate limits on installed capacity are therefore recommended.

Three DG sources were placed at buses 7, 8, and 9—the closest to the largest demand. Photovoltaic generation was chosen and parameterized after existing PV plants worldwide [4], taking the plants in Table I as a reference. The PV generation was modeled at unity power factor and a plant factor of 1, i.e., supplying only active power at full capacity.

TABLE I
REFERENCE PHOTOVOLTAIC PLANTS.

PV plant	Location	Installed capacity
El Altillo Solar Farm	Colombia	150 MW
Gansu Yumen Solar Farm	China	300 MW
Hebei Nangong Solar Farm	China	400 MW

Simulations with DG alone resolved the generator and line overloads, but the voltage remained below the acceptable range; the SVS was therefore added to inject the reactive power needed to raise the voltage. According to [5], an SVS installed at a bus that experienced a load change injects reactive power, raising the bus voltage while reducing the current drawn from higher-voltage buses and increasing that from lower-voltage buses, which lowers losses where the current is smaller and helps decongest the lines. Although PV plants can also inject reactive power, they are most commonly used for active-power injection, with the SVS providing the compensation.

Fig. 5 shows the resulting voltage profile after applying both solutions, and Fig. 6 shows the voltage angles, which now remain between -15° and 10° —a considerable reduction from the previous scenario.

III. ANALYSIS OF THE SIMULATION RESULTS

A. Comparison Between Scenarios

The bus voltages of the three scenarios are compared in Fig. 7. From Scenario 1 to Scenario 2 there is a voltage drop from bus 1 to bus 30 (with some exceptions); from bus 31 onward the three scenarios overlap, indicating that the first 30 buses are the ones affected by the load increase at bus 8. Bus 8 shows the largest drop, from 0.996 to 0.913 p.u. The change from Scenario 2 to Scenario 3 is even more pronounced: the voltage values nearly recover to those of Scenario 1. Comparing Scenario 1 and Scenario 3, the curves begin to overlap from bus 16 onward, and at bus 8 the SVS raised the voltage above nominal (from 0.996 to 1 p.u.); in general,

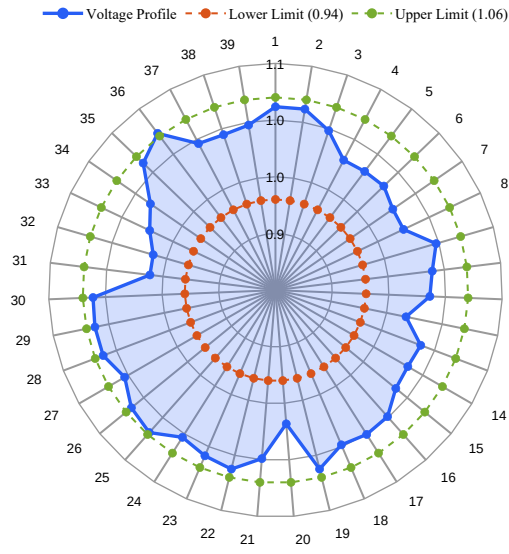


Fig. 5. Bus voltages of the EPS – Scenario 3.

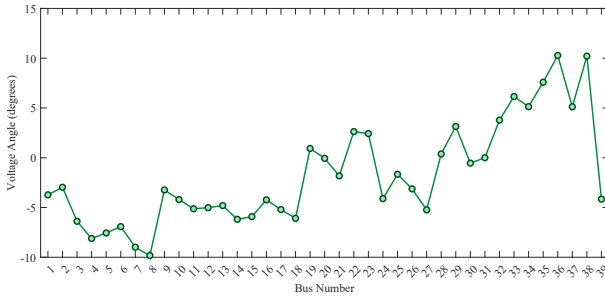


Fig. 6. Voltage angles at the EPS buses – Scenario 3.

at the buses where the curves do not overlap, the Scenario 3 voltages are slightly higher than those of Scenario 1.

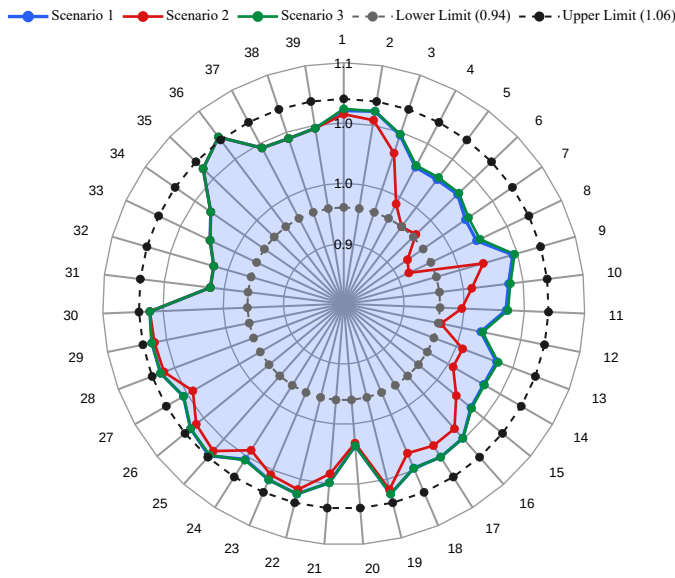


Fig. 7. Voltage comparison across the three scenarios.

Fig. 8 compares the voltage angles; as with the voltages, Scenario 2 is the most distant from the rest. The cause lies in the line power flows under the load increase and under the solutions. When the load increases and the system is not conditioned for it, bus 8 demands more power, so the generators must deliver more than in Scenario 1; this produces a larger angular opening at the buses in order to keep the system stable. Because bus 8 has no local generation, its load must be supplied by the flows of the lines connected to it: the larger the load, the larger those flows, which forces the angles to increase so that more power passes through the lines, altering the angles of the remaining buses and the line loadings.

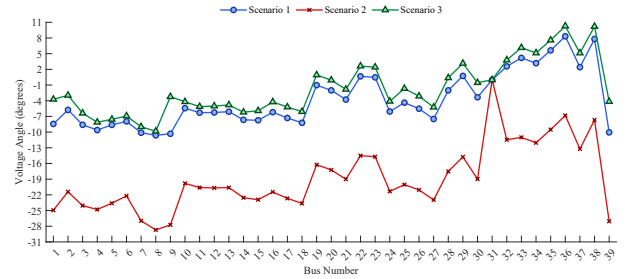


Fig. 8. Voltage-angle comparison across the three scenarios.

The opposite occurs with the DG and SVS. The PV generation was placed at buses 7, 8, and 9: buses 7 and 9 were chosen because, in Scenario 1, the lines between them and bus 8 are lightly loaded, so a controlled amount of power can be sent through them, while bus 8 was chosen to locally compensate part of the new load. The combination of DG and SVS stabilizes the voltage and reduces the need to transport power from other generation units, which lowers the power flows and the angular deviations; this is why the Scenario 3 angles are even smaller than those of Scenario 1.

Fig. 9 shows the active- and reactive-power dispatch of the units across the three scenarios. All generators except the reference unit dispatch the same active power, since they are voltage-controlled units; the reactive power shows the most variation. Scenario 2 is the most unstable, since the reference generator cannot compensate the new demand and must exceed its limits: with an active-power increase at the slack bus, the voltage angle rises, triggering a chain reaction to satisfy the power balance. To meet the flows and the new demand, the generators—with fixed active dispatch—increase their reactive dispatch up to their capability. The effect of the solution is visible in Scenario 3: the reference generator reduces its active dispatch and the other generators reduce their reactive dispatch, because the SVS compensates the reactive demand of the load and prevents the generators from overloading.

Fig. 10 shows the transmission-line loading in each scenario. In all three, line 27 is the most heavily loaded. This is because bus 22 mostly receives the dispatch of generator 6 (at bus 35), which delivers 650 MW and sends a current of 22–24 kA across the three scenarios; passing through the transformer, this reduces to about 1.064–1.084 kA, and all

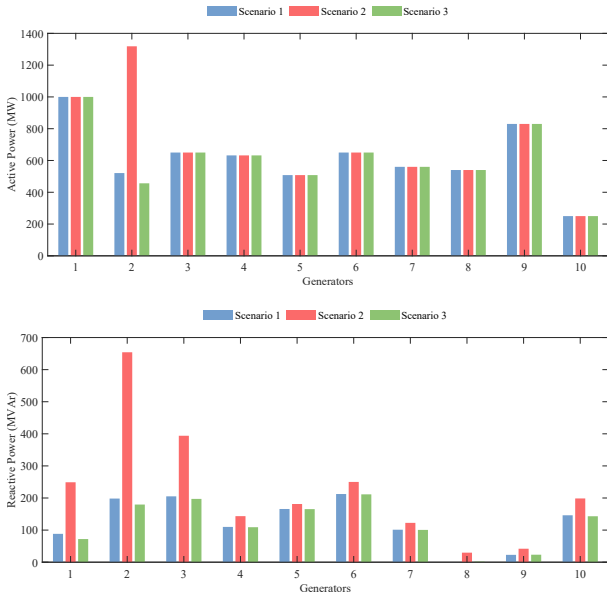


Fig. 9. Generator dispatch comparison across the three scenarios: active power (top) and reactive power (bottom).

lines are rated at 1 kA. Setting line 27 aside, Scenario 2 shows a considerable difference in the lines converging at bus 8 and nearby buses: the most loaded lines are those between buses 5, 6, 7, and 8. This confirms the angle analysis—with no local generation, the load must be supplied by the flows arriving at bus 8—and from line 22 onward the loadings begin to resemble one another across scenarios. Thus, in terms of angles the whole EPS is modified, whereas to cover the demand the generators and lines near bus 8 are prioritized, which is why the nearby lines are loaded beyond their nominal capacity.

Comparing Scenario 1 with Scenario 3, the line loadings do not follow a consistent pattern as the angles do, mainly because the solution injects active and reactive power into certain lines; from line 19 (bus 14–bus 15) onward the loadings begin to overlap. The effect of the solution is clearest between line 13 (bus 7–bus 8) and line 14 (bus 8–bus 9), whose loadings rise from 31.20% to 53.1% and from 18.1% to 56% respectively, these being the lines that carry the photovoltaic generation.

B. Comparison Between Scenario 1 and the PyPower Simulation

Finally, Scenario 1 is compared with the results of the PyPower library, used to analyze and solve power flows; the code developed in a Jupyter notebook for the 39-bus test system is also provided as an attachment. Fig. 11 shows the bus voltages from both tools, which coincide at most buses with only a few showing variation, and Fig. 12 shows the corresponding voltage angles.

The comparison reveals differing data at some buses, mainly regarding loads. Table II lists the differences found; a dash indicates that both tools use the same value.

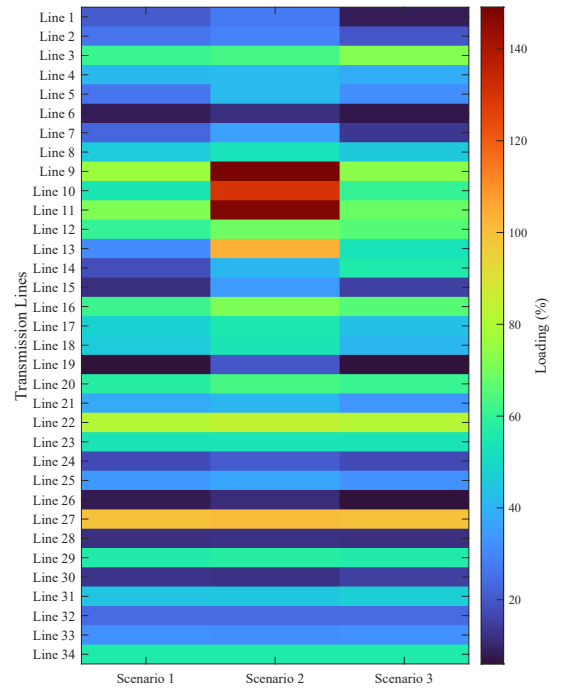


Fig. 10. Line-loading comparison across the three scenarios.

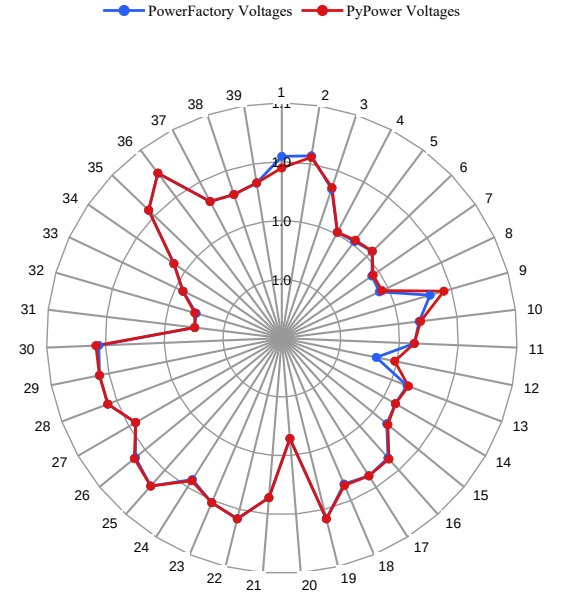


Fig. 11. Bus voltage comparison between tools: PowerFactory vs. PyPower.

These differences are one reason the two models produce different results: the PyPower model includes loads that the PowerFactory model does not account for, some of them with different active and reactive values. As seen in Scenario 2, changing a load value can substantially modify the whole network. The same happens here: because the PyPower model considers different values, they affect the rest of the EPS and its power flow. With different loads and active dispatch, the first quantities to change are the generators' reactive

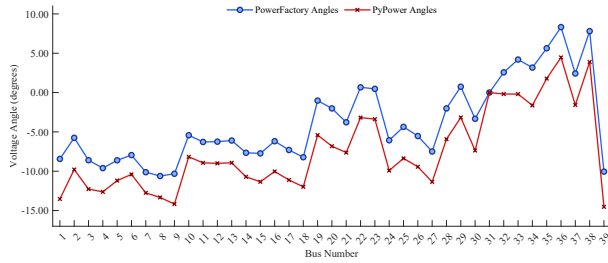


Fig. 12. Bus voltage-angle comparison between tools: PowerFactory vs. PyPower.

dispatch, and—as learned in the previous analyses—variations in generator dispatch cause variations in the line flows and a larger angular opening, as seen in Fig. 12. This is another reason the voltage results differ: the active and reactive power arriving at the buses, together with the line losses, differ between models, so some buses end up with different voltages while others, whose loads are the same in both models, remain similar.

TABLE II
COMPARISON OF LOAD AND GENERATOR DATA BETWEEN
POWERFACTORY AND PYPower.

Bus / Gen.	PowerFactory		PyPower	
	[MW]	[MVar]	[MW]	[MVar]
<i>Load demand</i>				
1	0	0	97.6	44.2
9	0	0	6.5	-66.6
12	7.5	—	8.53	—
20	628	—	680	—
<i>Generator dispatch</i>				
1	—	88.285	—	78.47
2	520.812	198.259	677.87	221.57
3	—	205.153	—	206.96
4	632	109.907	532	108.29
5	—	165.764	—	166.69
6	—	212.413	—	210.66
7	—	101.176	—	100.16
8	—	0.441	—	-1.37
9	—	22.842	—	21.73
10	—	146.16	—	161.76

IV. CONCLUSIONS

This work provided an understanding of power flows in an electric power system, specifically the IEEE 39-bus test system.

The normal-state analysis (Scenario 1) showed stable behavior under standard conditions, with voltages and angles within acceptable ranges, and served as the reference for evaluating the effects of any maneuver on the whole system. In Scenario 2, the load increase at bus 8 caused significant voltage drops at nearby buses and the saturation of some lines, confirming that power networks react to the smallest change: regardless of where the event occurs, the greatest disturbance appears locally and then propagates, altering the initial state. This was also confirmed by comparing the PowerFactory and

PyPower simulations, whose differences—mainly in loads and generators—altered the voltage and angle profiles at certain buses.

To correct the imbalance from the overload in Scenario 2, distributed generation and a static VAR compensator were implemented. The literature indicates that DG is effective for relieving line congestion and serving part of the local demand, while compensation devices such as the SVS improve voltage profiles; DG penetration, however, must be controlled and applied only when it is guaranteed not to harm the system. Together, these two solutions relieved the system—especially bus 8—redistributing the generator dispatch while correcting the voltages into an acceptable range.

Although DG alone could solve the Scenario 2 problem (since these generators can inject both active and reactive power), the SVS was added for academic purposes to validate the effectiveness of combining the technologies strategically. The PV generators were assumed to operate at unity power factor and full capacity, since the objective was not to model renewable generation in technical detail but to assess its positive impact on system stability. The comparison of the three scenarios showed favorable results for the Scenario 3 solutions: the losses, drops, and disturbances of Scenario 2 were efficiently mitigated, and the system recovered stability with voltages, angles, and dispatch similar to the initial state.

The work highlighted the importance of understanding power flows and of simulating the power system under different conditions, and it encouraged the development of creative, effective solutions—relevant when moving from academic work to fieldwork. Real power systems are dynamic and undergo abrupt, instantaneous changes; unlike simulations, which run in a static, controlled environment, real events require well-sized, resilient systems. This study showed that solutions such as distributed generation and compensation systems can increase system resilience, ensuring stable and safe operation. Finally, advanced monitoring tools such as WAMS and PMUs provide fast, accurate measurements that, integrated into specialized simulators, enable real-time analysis of different scenarios—crucial for anticipating and mitigating disturbances in the power system.

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